Find customers who have placed at least 5 orders.

Display:

CustomerID --> customers
Customer name --> customers
City --> customers
Number of orders --> orders

```
select
  c.CustomerID,
  c.Name,
  c.City,
  count(o.OrderID) as no_of_orders
from `e1.customers` as c
inner join `e1.orders` as o
on c.CustomerID = o.CustomerID
group by
  c.CustomerID,
  c.Name,
  c.City
having no_of_orders >= 5
order by no_of_orders desc
```

For every customer, number their orders chronologically.

Display:

CustomerID
OrderID
OrderDate
Order number

For example:

Customer C001
Order O001 → 1
Order O005 → 2
Order O010 → 3

```
select
  CustomerID,
  OrderID,
  OrderDate,
  row_number() over(partition by CustomerID order by OrderDate,
OrderID) as order_number
from `e1.orders`
order by CustomerID, order_number
```

```
Find the number of orders placed on Saturday and Sunday.

Display:

Day name
Number of orders



select format_date('%A', DATE '2026-09-10')

select extract(DAYOFWEEK from DATE '2026-09-10')




select
  format_date('%A', OrderDate) as day_name, --%A mean full day
name
  count(*) as order_count
from `e1.orders`
where extract(DAYOFWEEK from orderDate) in (1,7)
group by day_name
order by order_count desc




Classify every order as:
```

```
Weekend
Weekday

Then count the orders in each group.

select
  case
  when extract(DAYOFWEEK from orderDate) in (1,7) then 'Weekend'
  else 'Weekday'
  end as Day_type,
  count(*) as order_count
from `e1.orders`
group by Day_type




Find how many customers signed up in each calendar month,
irrespective of the year.

For example, combine all January signups together.


select
  extract(MONTH from signupdate) as signup_month,
  format_date('%B', signupdate) as Month_Name,
  count(*) as customer_count
from `e1.customers`
```

```
group by signup_month, Month_Name



Classify every order as:

First Half → January to June
Second Half → July to December

Then count orders.

select
  case
    when extract(MONTH from orderdate) <= 6 then 'First Half'
    else 'Second Half'
  end as year_type,
  count(*) as num_of_orders
from `e1.orders`
group by year_type




Calculate total sales for orders placed in the first half and
second half of the year.

select
   case
    when extract(MONTH from orderdate) <= 6 then 'First Half'
    else 'Second Half'
```

```
  end as year_type,
  sum(oi.total) as total_sales
from `e1.orders` as o
inner join `e1.order_items` as oi
  on o.OrderID = oi.OrderID
group by year_type
order by total_sales desc
```

Find which day of the week has the highest number of customer signups.

```
select
  extract(DAYOFWEEK from signupdate) as day_num,
  format_date('%A', signupdate) as day_name,
  count(*) as no_of_signup
from `e1.customers`
group by day_num, day_name
order by no_of_signup desc
```

Find how many orders were placed during the first seven days of each month.

```
select
  Date_trunc(OrderDate, MONTH) as order_month,
```

```
  count(*) as no_of_orders
from `e1.orders`
where extract(DAY from orderDate) <= 7
group by order_month




select
  Date_trunc(OrderDate, MONTH) as order_month,
  format_date('%B', OrderDate) as month_name,
  count(*) as no_of_orders
from `e1.orders`
where extract(DAY from orderDate) <= 7
group by order_month, month_name




Find customers who placed their first order in the
same calendar month and year in which they signed up.


with first_orders as (
  select
    CustomerID,
    min(orderdate) as first_order_date
  from `e1.orders`
  group by CustomerID
)
```

```sql
select
  c.CustomerID,
  c.Name,
  c.SignupDate,
  f.first_order_date
from `e1.customers` as c
inner join first_orders as f
on c.CustomerID = f.customerID
where
  date_trunc(c.SignupDate, MONTH) =
date_trunc(f.first_order_date, MONTH)
order by c.SignupDate
```

Calculate total sales for each calendar week.

Group customers into signup cohorts based on the month in which they registered.

Display:

Signup month
Number of customers